Lecture 03: First order linear ODE

Def^n: A differential equation is of the form $a_0(x)\frac{d^n y}{dx} + a_1(x)y = a_2(x)$ is said to be linear if

(i) The dependent variable y and its derivative are of the first deg degree; that is, the power of y and its derivatives is exactly 1.

(ii) The coefficients of y and its derivatives depend only on the independent variable x or constant.

Example: (1) $x\frac{dy}{dx} + 5y = \sin x$

(II) $10\frac{dy}{dx} + y\sin x = \cos x$

(III) $5\frac{dy}{dx} + 10y = 3$

(IV) $x^2\frac{dy}{dx} + y\tan x = e^x$

They are all linear ODE.

Example of non-linear ODE:

(i) $(1+y)\frac{dy}{dx} + 2y = e^x$

(ii) $\frac{d^2y}{dx} + x\sin y = 0$

(iii) $\frac{dy}{dx} + y^2 = x^2$

(iv) $\frac{d^2y}{dx} + x\tan y = \cos^2 y$

Solution of linear ODE:

$$a_0(x)\frac{dy}{dx} + a_1(x)y = a_2(x)$$

$$\Rightarrow \frac{dy}{dx} + \frac{a_1(x)}{a_0(x)}y = \frac{a_2(x)}{a_0(x)}$$

$$\Rightarrow \boxed{\frac{dy}{dx} + P(x)y = Q(x)} \quad (1)$$

which is the standard form of the linear ODE.

** At this stage, we will make the L.H.S of (1) is perfect differential, that is, directly integrable.

* Example: (i) $\int dx = x + C$, (ii) $\int d(x^2) = x^2 + C$

(iii) $\int d(NSU) = NSU + C$.

** We will multiply ODE 1 by a factor such that ODE 1 is reducible to integ-rable, that factor is known as integrating factor (I.F.).

Here

$$I.F. = e^{\int P(x)\,dx}$$

1 × I.F. ⇒

$$e^{\int P(x)\,dx}\frac{dy}{dx} + P(x)\,e^{\int P(x)\,dx}\,y = Q(x)\,e^{\int P(x)\,dx}$$

$$\Rightarrow \frac{d}{dx}\left(\text{I.F.} \times \text{dependent variable}\right) = Q(x)\,e^{\int P(x)\,dx}$$

$$\Rightarrow \frac{d}{dx}\left(e^{\int P(x)\,dx}\,y\right) = Q(x)\,e^{\int P(x)\,dx}$$

$$\Rightarrow d\left(e^{\int P(x)\,dx}\,y\right) = Q(x)\,e^{\int P(x)\,dx}\,dx$$

Integrating

$$\underbrace{e^{\int P(x)\,dx}\,y}_{\text{I.F} \times \text{dependent variable}} = \int Q(x)\,e^{\int P(x)\,dx}\,dx$$

(We have to apply Integration by parts)

Example 1: Solve $x\frac{dy}{dx} - 4y = x^6 e^x$

Ans: Given $x\frac{dy}{dx} - 4y = x^6 e^x$

$$\Rightarrow \frac{dy}{dx} - \frac{4}{x}y = x^5 e^x \quad \text{--- (1)}$$

Here $P(x) = -\frac{4}{x}$

$$\therefore \text{I.F.} = e^{-\int \frac{4}{x}dx} = e^{-4\ln x} = e^{\ln x^{-4}} = x^{-4} = \frac{1}{x^4}$$

$$(1) \times \text{I.F.} \Rightarrow \frac{1}{x^4}\frac{dy}{dx} - \frac{4}{x^5}y = xe^x$$

$$\Rightarrow \frac{d}{dx}\left(\frac{1}{x^4}y\right) = xe^x$$

$$\Rightarrow d\left(\frac{y}{x^4}\right) = xe^x\,dx$$

Integrating,

$$\frac{y}{x^4} = \int xe^x\,dx \quad \text{--- (2)}$$

Integrating by parts by tabular form

Rep. diff. of u		Repeated Integration of v
x	$+$	e^x
1	$-$	e^x
0		e^x

$= xe^x - e^x$

$$(2) \Rightarrow \quad \frac{y}{x^4} = xe^x - e^x + c$$

$$\Rightarrow \boxed{y = x^5 e^x - x^4 e^x + c\, x^4}$$

Ans

Example: Solve the IVP: $x\frac{dy}{dx} + y = 2x$, $y(1) = 0$

Ans: Given $x\frac{dy}{dx} + y = 2x$

$$\Rightarrow \frac{dy}{dx} + \frac{1}{x}y = 2 \quad \text{—— (1)}$$

$$P(x) = \frac{1}{x}$$

$$I.F. = e^{\int \frac{1}{x}dx} = e^{\ln x} = x$$

$$(1) \times I.F \Rightarrow \quad x\frac{dy}{dx} + y = 2x$$

$$\Rightarrow \frac{d}{dx}(xy) = 2x$$

$$\Rightarrow d(xy) = 2x\,dx$$

Integrating,

$$xy = \frac{\cancel{2}x^2}{\cancel{2}} + c$$

$$y = x + \frac{c}{x} \quad \text{—— (2)}$$

at $x=1,\ y=0$

$(2) \Rightarrow \quad 0 = 1 + \frac{c}{1}$

$\Rightarrow \quad c = -1$

(2) becomes. $\boxed{y = x - \frac{1}{x}}$

Example 3: Solve $\sin 2x \frac{dy}{dx} - y = \tan x$

Ans: The standard form of the given ODE is

$$\frac{dy}{dx} - y \operatorname{cosec} 2x = \frac{\tan x}{\sin 2x} = \frac{\tan x}{2 \sin x \cos x}$$

$$\Rightarrow \frac{dy}{dx} - y \operatorname{cosec} 2x = \frac{1}{2} \sec^2 x \quad \text{--- (1)}$$

Here $P(x) = -\operatorname{cosec} 2x$

$$I.F. = e^{-\int \operatorname{cosec} 2x \, dx} = e^{-\frac{1}{2} \ln |\tan x|}$$

$$= e^{\ln (\tan x)^{-1/2}} = \frac{1}{\sqrt{\tan x}}$$

$(1) \times I.F \Rightarrow \frac{1}{\sqrt{\tan x}} \frac{dy}{dx} - y \frac{\operatorname{cosec} 2x}{\sqrt{\tan x}} = \frac{1}{2} \frac{\sec^2 x}{\sqrt{\tan x}}$

$$\frac{d}{dx}\left(\frac{1}{\sqrt{\tan x}} y\right) = \frac{1}{2} \frac{\sec^2 x}{\sqrt{\tan x}}$$

Non-linear ODE: Bernoulli Equation

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \; ; \quad n \neq 0, 1$$

$$\Rightarrow y^{-n}\frac{dy}{dx} + P(x)y^{1-n} = Q(x) \quad \text{--- (1)}$$

Let $y^{1-n} = u$

$$(1-n)y^{-n}\frac{dy}{dx} = \frac{du}{dx}$$

(1) $\Rightarrow$ $\frac{1}{1-n}\frac{du}{dx} + P(x)u = Q(x)$ which is linear ODE

Example: Solve $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$

Ans: Given $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$

$$\Rightarrow y^{-2}\frac{dy}{dx} + \frac{1}{x}y^{-1} = \frac{1}{x^2} \quad \text{--- (1)}$$

Let $y^{-1} = u$

$$\therefore -y^{-2}\frac{dy}{dx} = \frac{du}{dx}$$

P.T.O.

(1) $\Rightarrow -\frac{du}{dx} + \frac{1}{x} u = \frac{1}{x^2}$

$\Rightarrow \frac{du}{dx} - \frac{1}{x} u = -\frac{1}{x^2}$ — (2)

$I.F. = e^{-\int \frac{1}{x} dx}$ Here $P(x) = -\frac{1}{x}$

$= e^{-\ln|x|} = e^{\ln x^{-1}} = \frac{1}{x}$

(2) $\times$ I.F. $\Rightarrow \frac{1}{x}\frac{du}{dx} - \frac{1}{x^2} u = -\frac{1}{x^3}$

$\Rightarrow \frac{d}{dx}\left(\frac{1}{x} u\right) = -\frac{1}{x^3}$

$\Rightarrow d\left(\frac{u}{x}\right) = -\frac{1}{x^3} dx$

$\Rightarrow \frac{u}{x} = -\frac{1}{2x^2} + C$

$\Rightarrow u = \frac{1}{2x} + Cx$

$\Rightarrow \boxed{\frac{1}{y} = \frac{1}{2x} + Cx}$

Example:
Solve $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$

A: $\sec^2 y \frac{dy}{dx} + x \frac{2 \sin y \cos y}{\cos^2 y} = x^3$

$\Rightarrow \sec^2 y \frac{dy}{dx} + 2x \tan y = x^3$ — (1)

Let $\tan y = u \therefore \sec^2 u \frac{dy}{dx} = \frac{du}{dx}$

(1) $\Rightarrow \frac{du}{dx} + 2xu = x^3$ —— (2)

$I.F. = e^{\int 2x\,dx} = e^{x^2}$

(2) $\times$ I.F $\Rightarrow e^{x^2}\frac{du}{dx} + 2xe^{x^2} = x^3 e^{x^2}$

$\Rightarrow \frac{d}{dx}(e^{x^2}u) = x^3 e^{x^2}$

$\Rightarrow d(e^{x^2}u) = x^3 e^{x^2}\,dx$

Integrating $e^{x^2}u = \int x^3 e^{x^2}\,dx$ —— (3)

Let $x^2 = z$

$2x\,dx = dz$

<u>Here</u> $\int x\,x^2 e^{x^2}\,dx$

$= \frac{1}{2}\int z\,e^{z}\,dz$

$= \frac{1}{2} z\,e^{z} - e^{z}$

$= \frac{1}{2} x^2 e^{x^2} - e^{x^2}$

(3) $\Rightarrow e^{x^2}.\tan y = \frac{1}{2}x^2 e^{x^2} - e^{x^2} + C$

Home task: Linear ODE

(1) $x \ln \frac{dy}{dx} + y = 2 \ln x$

(2) $(1+y^2)\, dx = (\tan^{-1} y - x)\, dy$; Hints, let x dependent variable.

Gillis book: Page 61

Q, 12, 13, 14, 16, 17, 18, 21,

IVP: 33, 34,

Non-linear: (1) $\frac{dy}{dx} + \frac{x}{1-x^2} y = x\sqrt{y}$

(2) $(1-x^2)\frac{dy}{dx} + xy = xy^2$; Hints $I.F = \sqrt{1-x^2}$.

(3) $\frac{dy}{dx} + (2x\tan^{-1} y - x^3)(1+y^2) = 0$

(4) $x\frac{dy}{dx} + y = (xy)^{3/2}$; $y(1) = 4$.